Dancing Hexapod

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We are building a assistive mobility hexapod for AXIS Dance Company

Robot used in this case is from 2025 Mechatronic Design team

Design requirements

$1000 budget for assembly
$2500 budget for motors, drivers, encoder, battery

Weigh less than 150 lb, while carrying more than 170lb
No discernible damage when dropping 25 lb loads on top

Legs must drive locomotion (aux. Wheels allowed support only, no power/odom)

Ability to move forward, backward, and turn either direction with zero radius

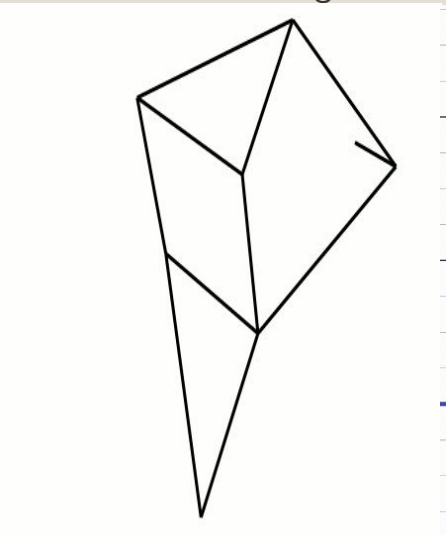
At least 30 min battery life in worst-case motion

Maximum: 30 seconds for 10’ forward motion
45 seconds for 10’ forward curvilinear - unknown angle!
45 seconds for 10’ backward motion
60 seconds for 10’ backward curvilinear - unknown angle!

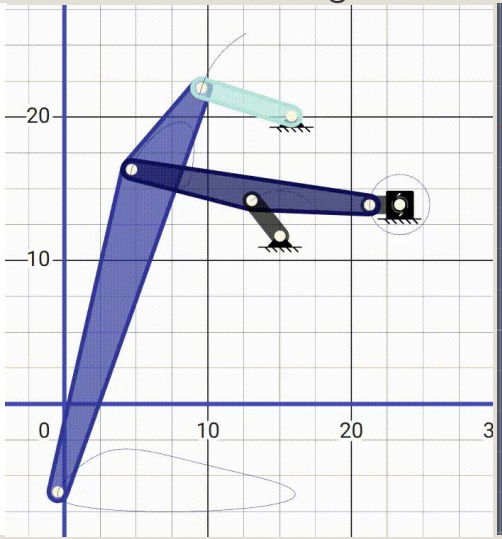
Goal: Be faster than the other (3) mechatronic teams!

System depiction - Klann Linkage Style Leg

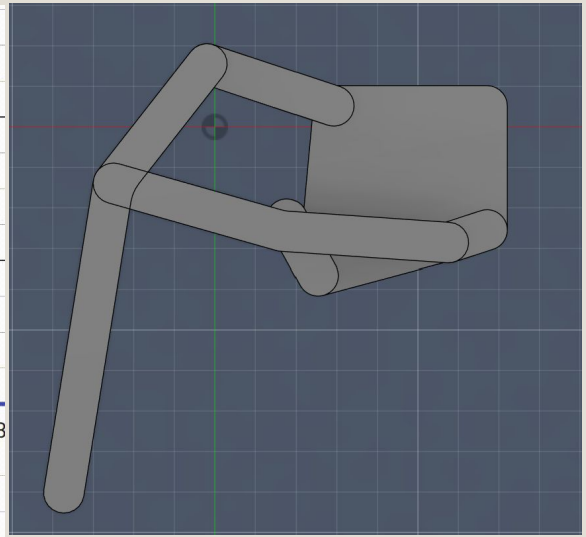
Jensen Linkage



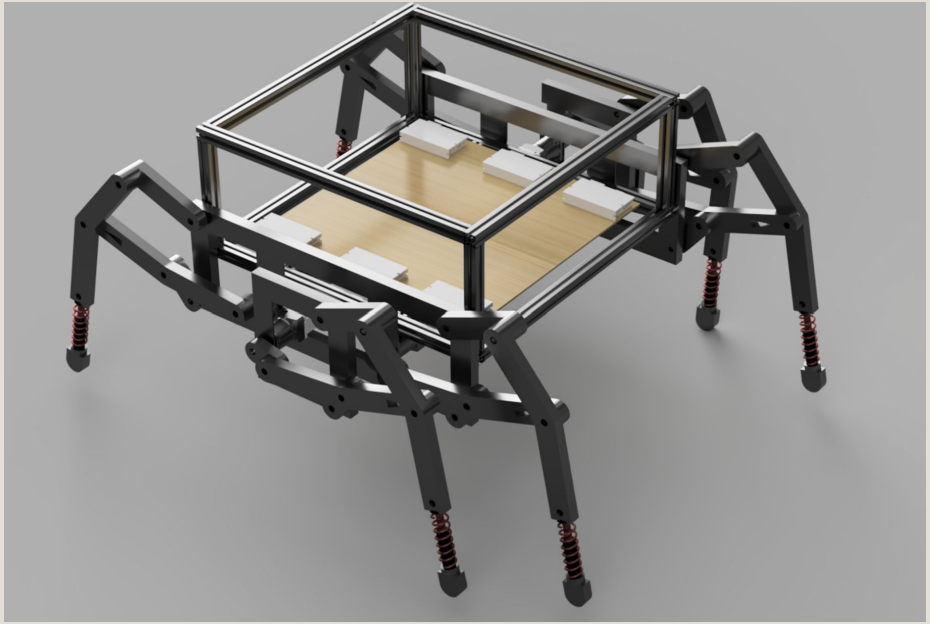
Klann Linkage

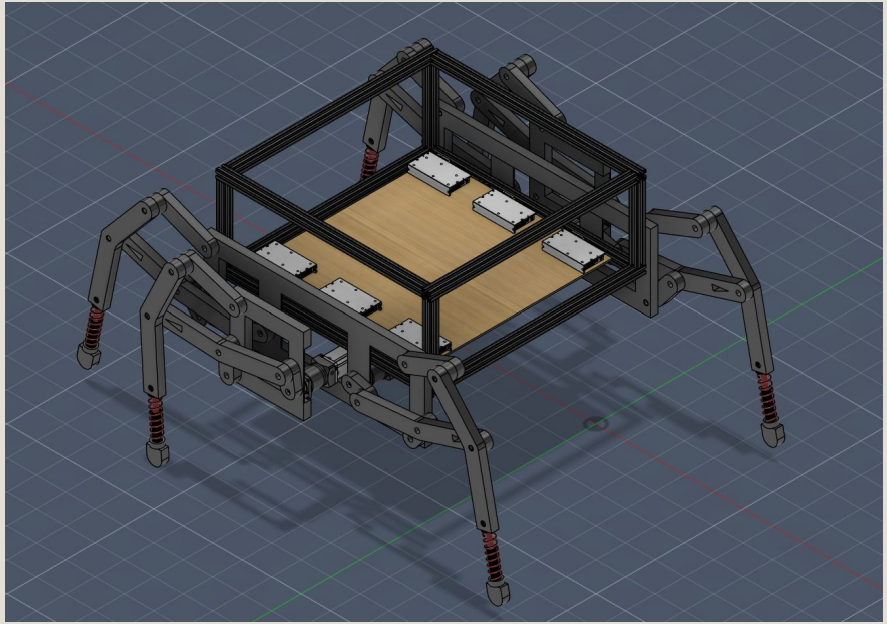


CAD Model

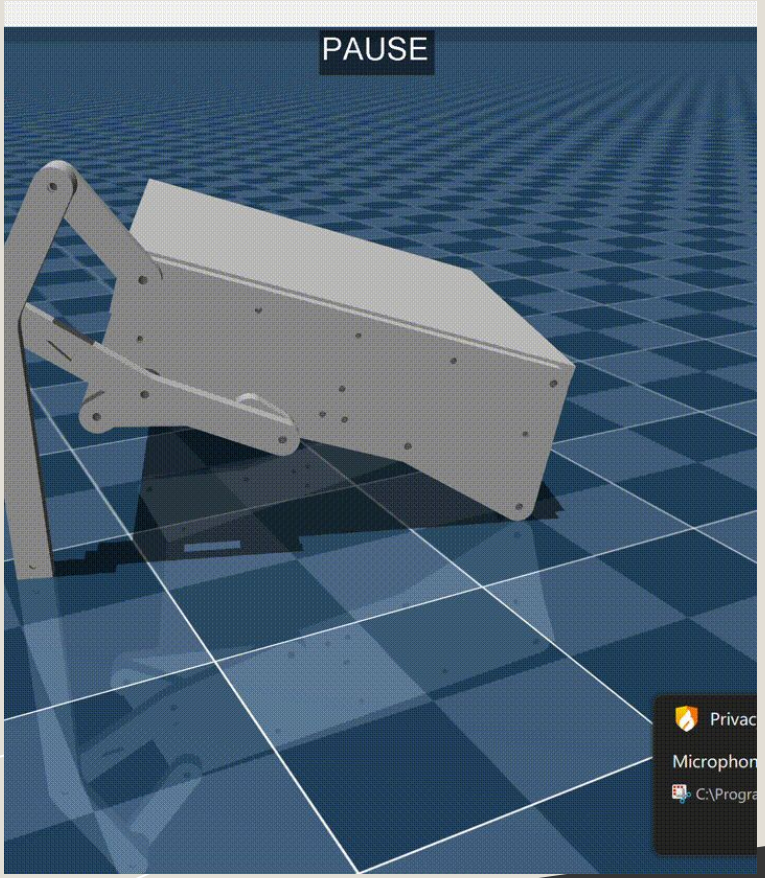


CAD DESIGN

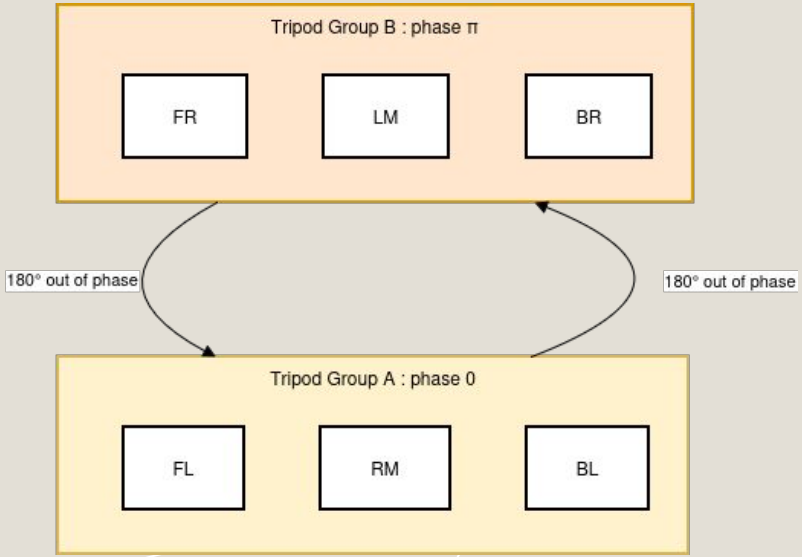


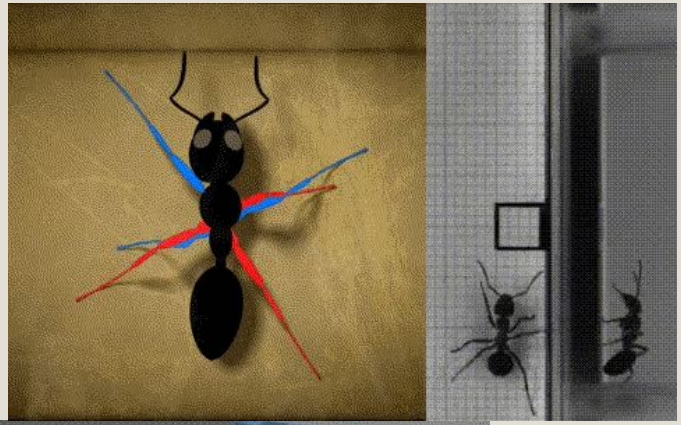


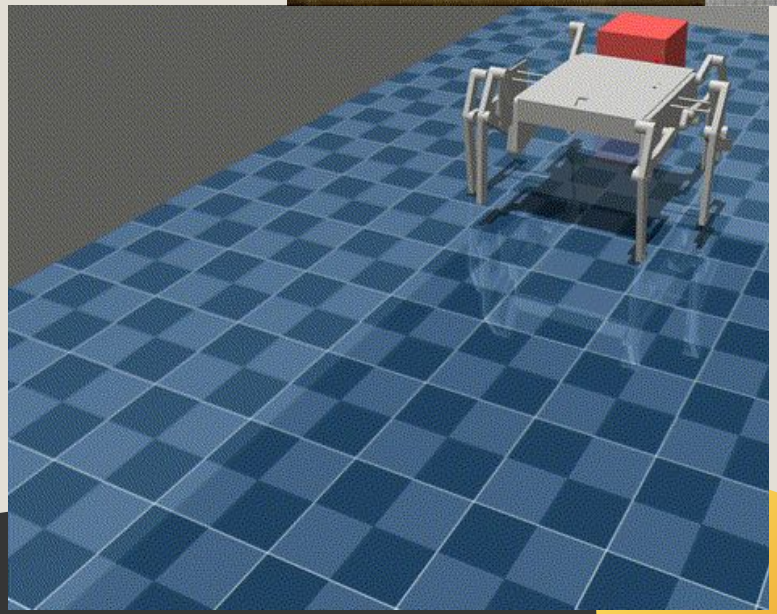
Simulating closed loop kinematics proved to be difficult



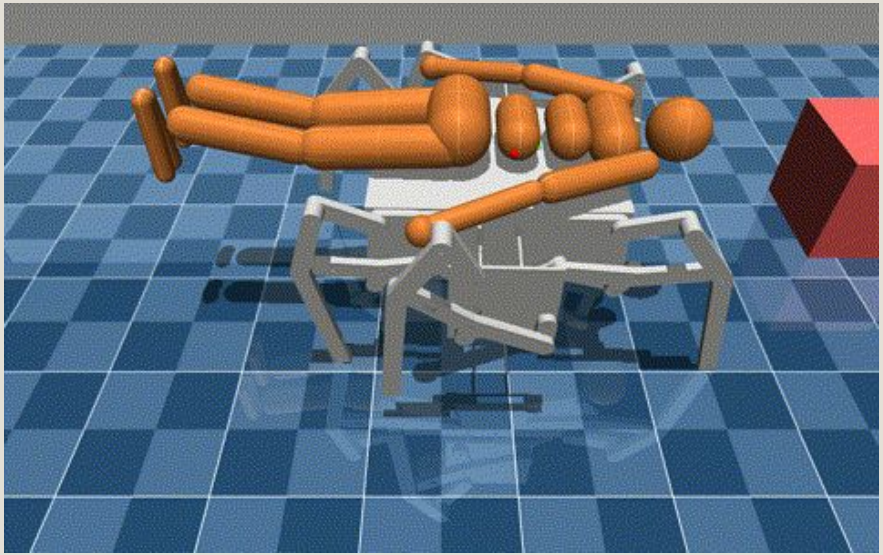
CPG phase control for insect-like stable gait
Achieves reasonable untuned stability.



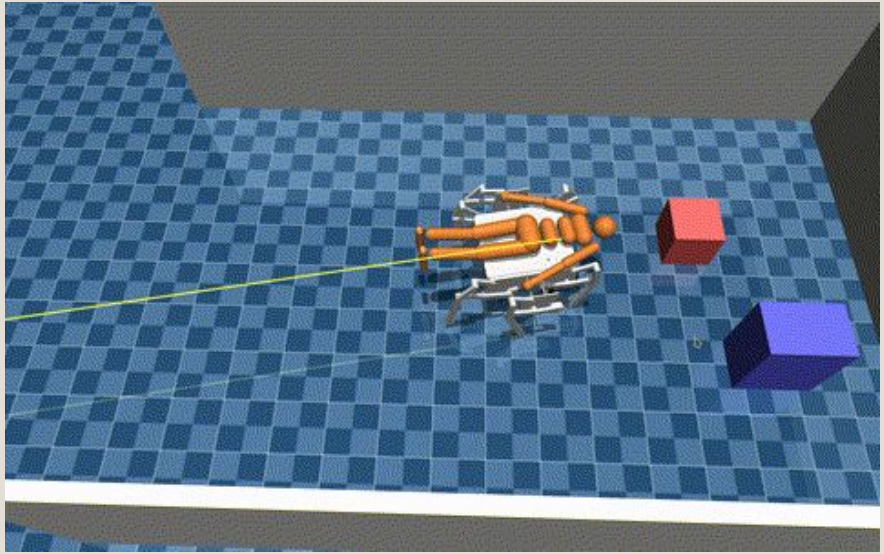




How can we carry variable loads with stability on our hexapod?

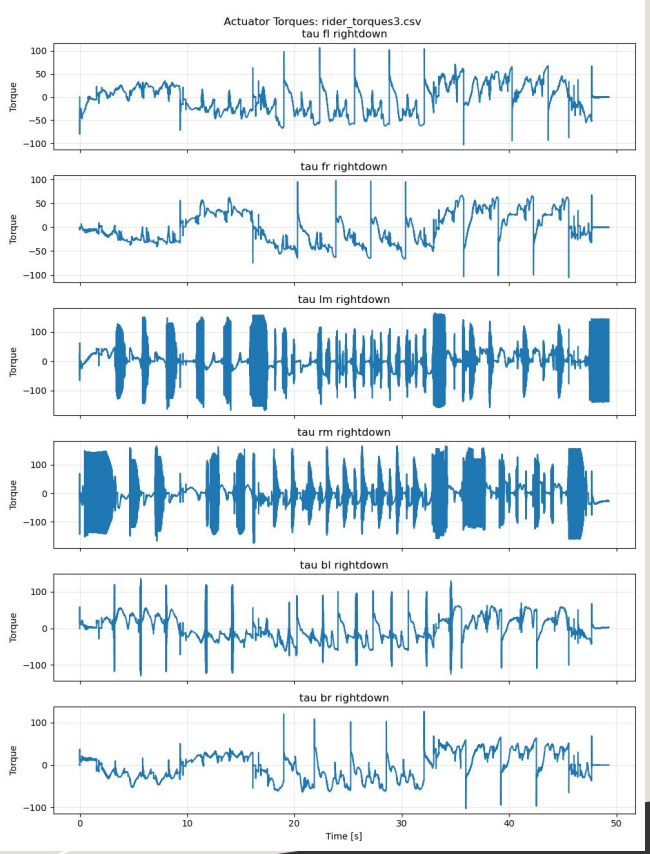


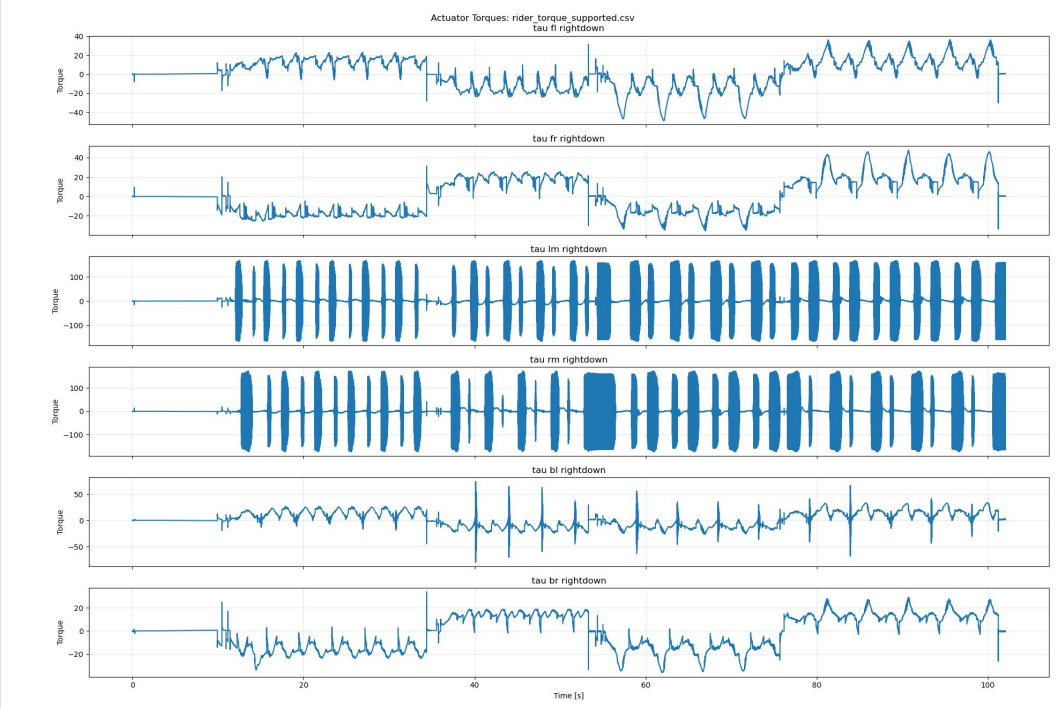
Unsupported



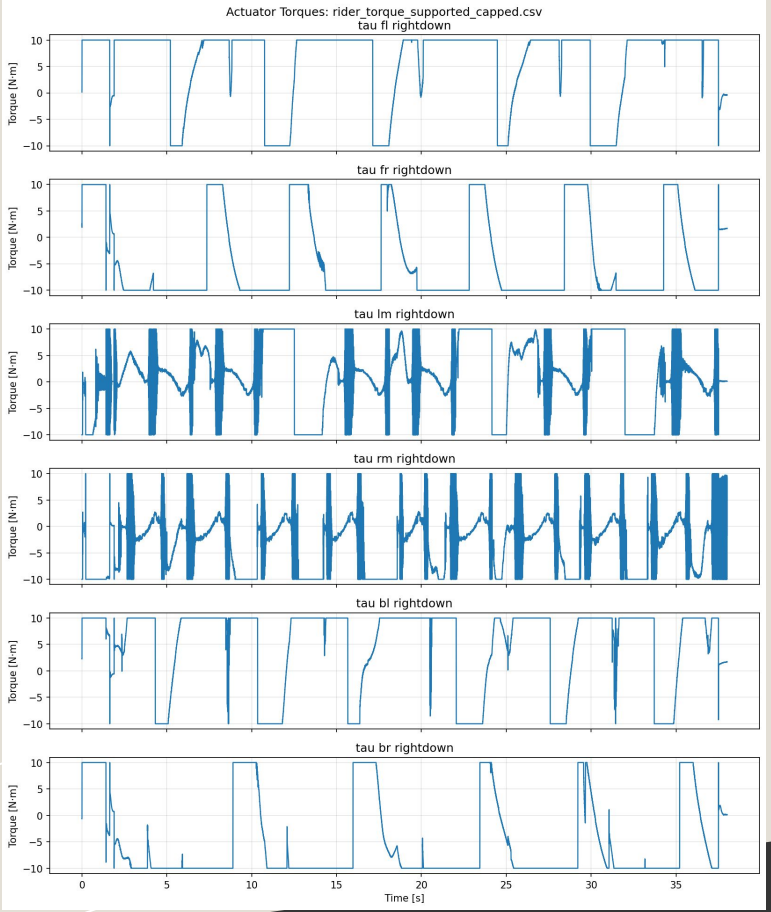
Supported

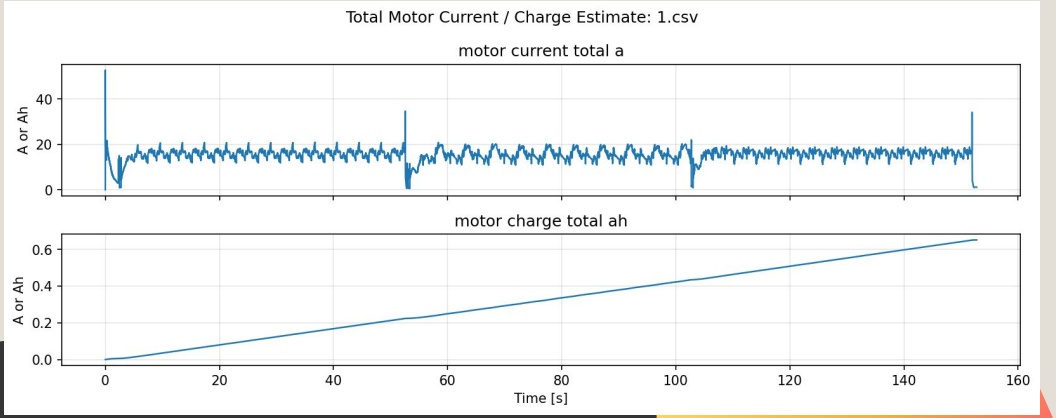
Torque-unconstrained (or torque naïve) control demands unrealistic output regardless of central support



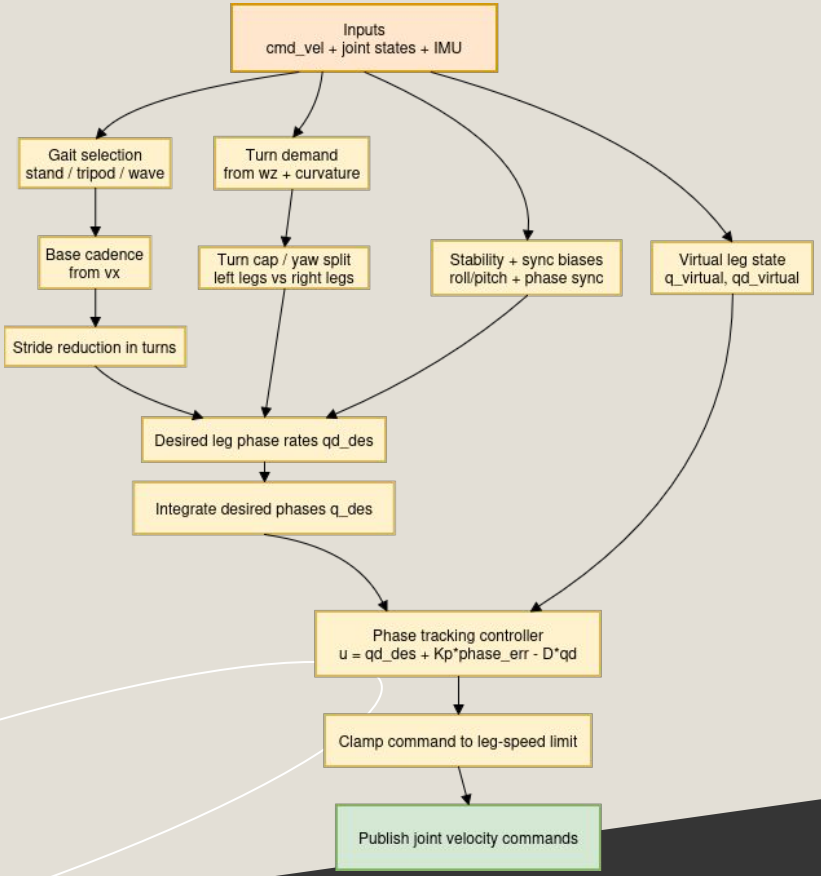


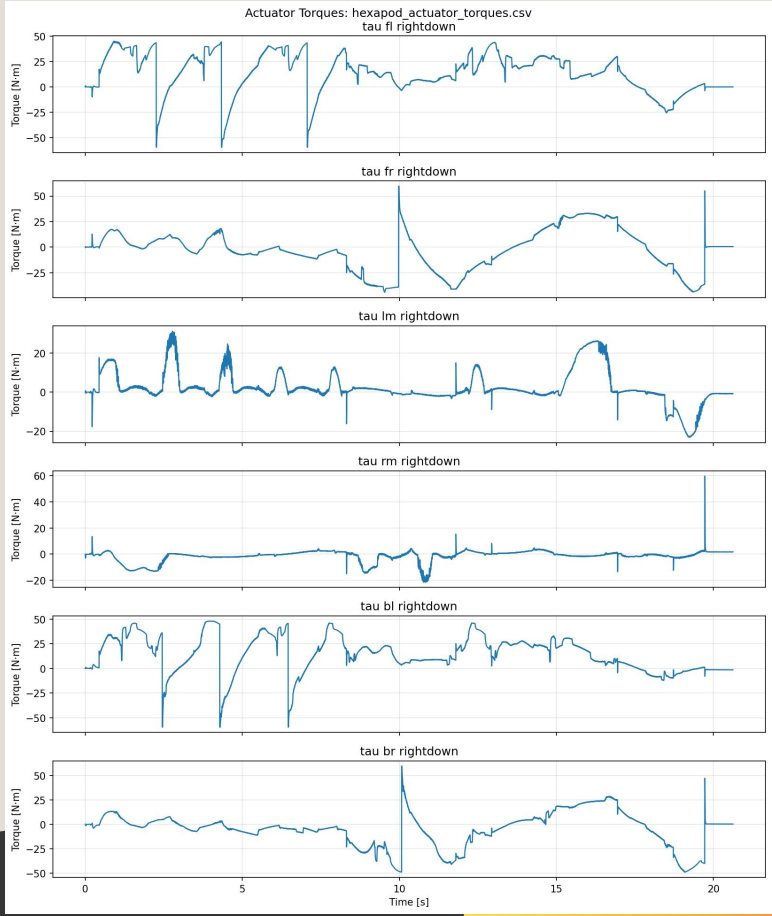
CPG struggles with torque constraints coming from weight/budget constraints





FSM-based impedance control allows explicit factoring of step size, and strength of contact with the ground





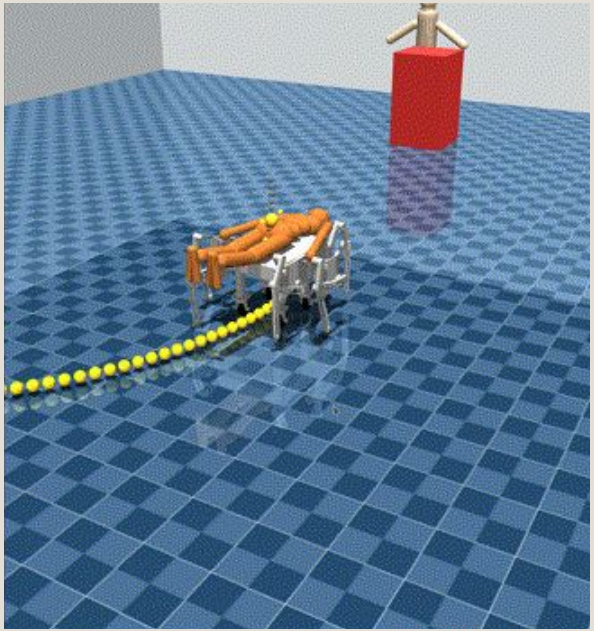
Ongoing steps - gait weight tuning run

Set up mjx environment to tune parameters inside the FSM controller.
Use exact specification requirements to achieve lower-effort arc following

Add touch sensing explicitly in physical and simulated environment.

Move towards a proper local planner with costmaps for motion when we do not have a known perfect path.

Implement RGBD SLAM to populate environment to plan through.

Fabricate first full-leg assembly to confirm kinematics.

Code embedded device drivers to drive physical servo motors.